WORKING DOCUMENT · DESIGN INPUT FOR THE BUILD

Sales-drop scenarios and detection rules

Every reason a SKU can miss its target, how the system detects each one from the six tables, and which columns are needed to build the hypothesis.

> **The one rule this document exists to serve**
> Every hypothesis is built in code, from columns, before the model is called. The model receives a finished set of findings and explains them. It never sees the raw tables and never works out the cause for itself. That is what makes the answer repeatable and checkable.

How to read this

SECTION	WHAT IT COVERS
Detection order	Which checks run first, and why the order changes the answer.
Summary	All 17 scenarios on one page.
Groups A to G	One block per scenario. Manager view, real cause, columns, test, whether it produces a number, and the document that confirms it.
Evidence contract	The exact object handed to the model once the checks have run.
Build list	Which scenarios to implement first for the demo.

"Gives a number?" is the important column. Some causes can be measured exactly. Some can only be pointed at. Mixing the two is how a system starts inventing percentages.

Detection order

The checks do not all carry the same weight. Some cancel the investigation entirely. Running them in the wrong order produces a confident answer to the wrong question.

ORDER	GROUP	WHY IT RUNS HERE
1	A · Context checks	These can stop everything. If the product is being discontinued on purpose, or the whole department is down by the same amount, there is no SKU problem to investigate.
2	B · Supply	Ask "could the customer buy it?" before asking "did they want it?" A stock-out explains a drop completely and needs no further theory.
3	C · Price and promotion	Price effects are exact arithmetic. Measure them before reaching for softer explanations.
4	D, E, F · Range, quality, channel	Narrower causes. Each explains a specific shape of drop.
5	G · Unexplained	Whatever is left. Only if this is large does the system look outside the business.

Why order matters more than accuracy
Imagine a SKU down 30%. The department is also down 28%. If the supply check runs first it will find a small stock issue and report it as the cause. The manager fixes stock and the problem stays, because the real story was that the whole department softened. Group A runs first so the system answers the right question.

All scenarios at a glance

ID	SCENARIO	MAIN TABLES USED	NUMBER?
A1	Product being discontinued	`products`	No — stops the investigation
A2	Whole category is down	`sales, targets`	Yes
A3	Previous period was a spike	`sales`	Yes
A4	Product too new to compare	`products`	No
A5	Target was set too high	`targets, sales`	Yes
B1	Complete stock-out	`stock, sales`	Yes
B2	Low stock, never zero	`stock, sales`	Yes
B3	Broken size curve	`products, stock, sales`	Yes
B4	Delivery arrived late	`receipts, stock`	Yes
B5	Reorder never placed	`receipts, stock`	Yes
C1	Discount withdrawn	`sales`	Yes
C2	Price increased	`sales, products`	Yes
C3	Campaign ended	`sales, stock`	No — residual only
D1	Cannibalised by sibling SKU	`sales, products`	Yes
D2	Lost visibility to a new launch	`sales, products`	No
E1	Returns spike	`returns, sales`	Yes
F1	One channel only	`sales`	Yes
G1	Genuinely unexplained	`all of the above`	Yes — the residual

Eighteen rows. Fourteen produce a measurable number. Four can only be pointed at, and those four must never be given a percentage.

Group A · Context checks

These run first. Each one can cancel the investigation or change what the question means.

A1 · Product is being discontinued

FIELD	DETAIL
Manager sees	A SKU declining week after week, with discount creeping up.
Real cause	Nothing is wrong. The product is being wound down on purpose.
Columns needed	`products.sku · products.status · sales.discount_pct`
Test in code	status is **clearance** or **discontinued**.
Gives a number?	No. The investigation stops here and the system says the decline is intentional.
Document that confirms	Range review meeting notes.
Watch out	If this check does not run first, the system will hunt for a cause that does not exist and give a confident wrong answer.

A2 · The whole category is down, not this SKU

FIELD	DETAIL
Manager sees	This SKU missed its target by 30%.
Real cause	The department is down 28%. The SKU is behaving almost exactly like everything around it.
Columns needed	`sales.units · sales.revenue · sales.week · products.model · products.department · targets.target_revenue`
Test in code	Work out the gap against target at three levels: SKU, model, department. If the SKU gap and the department gap are within a few points, the SKU is not the story.
Gives a number?	Yes. "SKU down 30%, department down 28%, so the SKU-specific effect is only 2%."
Document that confirms	Monthly trading meeting notes.
Watch out	This is the check that stops managers fixing the wrong thing. It should be visible in every answer, not only when it triggers.

A3 · The previous period was unusually high

FIELD	DETAIL
Manager sees	Sales fell 40% compared with last month.
Real cause	Last month was a promotional spike. This month is normal.
Columns needed	`sales.units · sales.week (needs 12 or more prior weeks)`
Test in code	Compare the previous period against its own trailing 12-week average. Flag if it was more than about 1.4 times that average.
Gives a number?	Yes. Split the fall into "return to normal" and "real decline".
Document that confirms	Campaign plan, if the spike was a promotion.

A4 · Product is too new to compare

FIELD	DETAIL
Manager sees	A new SKU is behind target.
Real cause	It launched six weeks ago. There is no trend to compare against and the target was a guess.
Columns needed	`products.launch_date · sales.week`
Test in code	Launch date is inside the last 8 weeks.
Gives a number?	No. The system suppresses trend and year-on-year claims and says why.
Document that confirms	Range review or campaign brief.

A5 · The target was set too high

FIELD	DETAIL
Manager sees	A SKU misses target every single month.
Real cause	The target assumed growth the product has never achieved.
Columns needed	`targets.target_units · targets.month · sales.units (52 weeks) · targets.source_document`
Test in code	Compare the target against the recent run rate. Flag if the target implies growth well above the trend of the last three months.
Gives a number?	Yes. Split the gap into "target stretch" and "actual decline".
Document that confirms	Monthly trading meeting notes, which is where the target was set.
Watch out	This is politically awkward and factually useful. State it plainly and cite the meeting.

Group B · Supply

Ask whether the customer could buy it before asking whether they wanted it.

B1 · Complete stock-out

FIELD	DETAIL
Manager sees	Units collapsed.
Real cause	There was nothing on the shelf for part of the period.
Columns needed	`stock.week · stock.sku · stock.units_on_hand · sales.units (trailing 8 weeks for the normal rate of sale)`
Test in code	Find weeks in the period where units_on_hand reached zero, or fell below half a week of normal demand.
Gives a number?	Yes. Lost units = normal weekly demand minus units actually sold, for each affected week.
Document that confirms	Buying committee minutes, or the supply chain report.
Watch out	Demand that could not be served is never observed. State the assumption used to estimate it.

B2 · Low stock, but never zero

FIELD	DETAIL
Manager sees	Stock reports look healthy. Sales are still down.
Real cause	Stock never hit zero but was too thin to serve a full week of demand.
Columns needed	`stock.units_on_hand · sales.units (trailing average)`
Test in code	weeks_cover = units_on_hand divided by average weekly units. Flag any week below 1.0 even though stock was above zero.
Gives a number?	Yes, partially. Constrained demand for those weeks.
Document that confirms	Supply chain report.
Watch out	This is the case a normal dashboard misses completely. "Stock: 40 units" looks fine until you know the product sells 95 a week.

B3 · Broken size curve

FIELD	DETAIL
Manager sees	Total stock is healthy. Sales keep falling.
Real cause	The sizes people actually buy sold out. What is left is the sizes nobody wants.
Columns needed	`products.size · products.model · stock.units_on_hand (per SKU) · sales.units (52 weeks, per SKU, for the normal size mix)`
Test in code	For each size, compare its share of current stock against its share of sales over the last year. Flag when the top-selling sizes have less than one week of cover.
Gives a number?	Yes.
Document that confirms	Buying committee minutes.
Watch out	Footwear only. This is the single best example of a problem that is invisible at model level and obvious at SKU level.

B4 · Delivery arrived late

FIELD	DETAIL
Manager sees	A stock-out, even though a reorder was placed.
Real cause	The supplier was late.
Columns needed	`receipts.sku · receipts.expected_date · receipts.actual_date · receipts.units_received · stock.units_on_hand`
Test in code	A receipt exists where actual date is more than a week after expected date, and a stock-out happened in that gap.
Gives a number?	Yes, through the stock-out calculation in B1.
Document that confirms	Supply chain status report.
Watch out	Different cause from B5 and a completely different fix. Do not merge them.

B5 · Reorder was never placed

FIELD	DETAIL
Manager sees	A stock-out.
Real cause	Somebody decided not to reorder. Usually to free up budget for something else.
Columns needed	`receipts (absence of any row for the SKU) · stock.units_on_hand (falling to zero)`
Test in code	Stock fell to zero and there is no receipt row in the preceding 8 weeks.
Gives a number?	Yes, through B1.
Document that confirms	Buying committee minutes — the decision, with a date and a reason.
Watch out	This is the demo scenario. The number comes from the tables; the reason comes from a meeting note; and the two line up on a date.

Group C · Price and promotion

Two of these are exact arithmetic. The third can never be given a number, and that distinction matters more than anything else in this document.

C1 · Discount withdrawn

FIELD	DETAIL
Manager sees	Sales dropped sharply. Nothing else changed.
Real cause	The product was discounted last period and is now at full price.
Columns needed	`sales.discount_pct (this period and prior) · sales.units · sales.revenue`
Test in code	Average discount fell from above zero to near zero between the periods.
Gives a number?	Yes. Exact price effect: the change in realised price multiplied by units.
Document that confirms	Campaign plan.

C2 · Price increased

FIELD	DETAIL
Manager sees	Revenue held up. Units fell.
Real cause	The list price went up.
Columns needed	`sales.revenue · sales.units (to get realised price) · products.price · sales.discount_pct`
Test in code	Realised price, which is revenue divided by units, rose while discount stayed flat.
Gives a number?	Yes. Exact price effect.
Document that confirms	Range review or trading meeting.

C3 · Campaign ended

FIELD	DETAIL
Manager sees	Sales fell off a cliff from one week to the next. Stock was fine. Price did not change.
Real cause	The promotional support stopped. Nothing about the product changed.
Columns needed	`sales.units · sales.discount_pct (unchanged) · stock.units_on_hand (adequate)`
Test in code	Units fell sharply week on week while stock was adequate and discount was unchanged. The system cannot name the campaign from the tables — it only knows the shape.
Gives a number?	**No.** To say "the campaign caused 24%" you would need to know what sales would have been if the campaign had never ended. Nothing in the data tells you that.
Document that confirms	Campaign plan, which supplies the end date and the featured SKU list.
Watch out	This is the trap. The system reports the residual as a measured number, and cites the campaign as the documented explanation for that residual. It never splits a percentage out for the campaign itself.

> **Measured effects get percentages. Documented effects get timing.**
> B1, B2, C1 and C2 produce exact numbers because they are subtraction. C3 produces no number of its own — it explains a number that was already measured. Written properly: *"39% of the fall is units that sat on the shelf at full price. That decline starts the week the Mid-Year Sale ended, and the campaign plan lists this SKU as featured."* The percentage is arithmetic. The reason is a citation. Nothing in between was invented.

Group D · Range and lifecycle

D1 · Cannibalised by a sibling SKU

FIELD	DETAIL
Manager sees	This SKU is down. The manager assumes demand has fallen.
Real cause	A sibling colour or a newer model in the same range absorbed the demand. Total sales are unchanged.
Columns needed	`sales.units (per SKU) · products.model · products.department · targets`
Test in code	This SKU is down, another SKU in the same model is up by a similar number of units, and the model total is close to target.
Gives a number?	Yes. Units moved between SKUs.
Document that confirms	Range review or campaign brief.
Watch out	Reporting this as a SKU failure would be actively misleading. Nothing was lost.

D2 · Lost visibility to a new launch

FIELD	DETAIL
Manager sees	A steady product suddenly slowed. No stock or price change.
Real cause	A new range launched and took the homepage and the marketing budget.
Columns needed	`sales.units · products.launch_date (other SKUs in the same department)`
Test in code	One or more new SKUs launched in the same department during the period.
Gives a number?	No. Circumstantial, and must be labelled as such.
Document that confirms	Autumn drop campaign brief.

Group E · Quality

E1 · Returns spike

FIELD	DETAIL
Manager sees	Net sales are weak. Gross sales look fine.
Real cause	The product is selling and coming back. Sizing, a defect, or a colour that looks different online.
Columns needed	`returns.units_returned · returns.reason · returns.week · sales.units`
Test in code	Return rate, which is units returned divided by units sold, is well above the trailing baseline for that SKU.
Gives a number?	Yes. Units returned is a counted number.
Document that confirms	Returns and quality summary.
Watch out	Only visible because returns are a separate table. If returns were netted into sales this would be invisible.

Group F · Channel

F1 · Drop in one channel only

FIELD	DETAIL
Manager sees	Total sales down.
Real cause	Stores are flat and e-commerce collapsed, or the reverse. A site problem, or a store closed for refit.
Columns needed	`sales.channel · sales.units · sales.revenue · sales.week`
Test in code	Work out the gap against target separately for each channel. Flag when one channel is far worse than the other.
Gives a number?	Yes.
Document that confirms	Store and e-commerce operations report.
Watch out	Always check this. A channel-specific cause and a demand problem look identical at total level.

Group G · Unexplained

G1 · Genuinely unexplained

FIELD	DETAIL
Manager sees	A real drop with no internal explanation.
Real cause	Unknown. Possibly something outside the business.
Columns needed	`the residual left after all checks above`
Test in code	Add up everything the measured checks explain. If more than about 30% of the gap is still unexplained, open the external gate.
Gives a number?	Yes. The residual itself is a measured number.
Document that confirms	External news and competitor watch, at lower confidence.
Watch out	The system must be allowed to end here and say it does not know. A system that always finds an explanation is inventing some of them.

What the model actually receives

Once every check has run, the result is one object. This is the only thing the model sees. It never receives the sales table.

```
question      "Why did MRL-CB-TAN drop last month?"
resolved     sku=MRL-CB-TAN  weeks=22..26  vs=June target

gap          target 480 units / actual 214 units / -266 units (-55%)

context      discontinued        no
             department_gap      -4%      (SKU gap -55%)
             prior_period_spike  no
             product_age_weeks   61
             target_stretch      none

findings     [ id=B5  label="reorder never placed"
                 units=-161  share=0.61  confidence=high
                 basis="stock 0 in wk24-26; no receipt in 8 weeks"
                 evidence=[stock rows, receipts absence] ]

             [ id=C3  label="campaign ended"
                 units=-105  share=0.39  confidence=medium
                 quantified=false
                 basis="units fell wk27 with stock ok, discount 0"
                 evidence=[] ]

residual     units=0  share=0.00        -> external gate CLOSED

documents    [ Buying_Committee_2026-05-12.pdf p3  matched=B5 ]
             [ MidYear_Campaign_Plan_SS26.docx p1  matched=C3 ]

external     not searched (residual below threshold)
```

Why this shape
Every number is already calculated. Shares already add to 1.0. Each finding carries the basis it was derived from and a confidence level. Findings marked **quantified=false** tell the model it may describe the cause but must not attach a percentage to it. The model writes prose around this object. It cannot add a cause that is not in the list, and it has no arithmetic to do.

What to build first

Seventeen scenarios is the full map. The demo needs five, chosen to cover different shapes of cause and to exercise both the gate open and the gate closed.

PRIORITY	SCENARIO	WHY THIS ONE
1 · hero	B5 reorder never placed + C3 campaign ended	Two causes in one answer. One measured exactly, one only pointed at. The gate stays closed and the system says it did not need to look outside. This is the scenario that shows the whole design.
2	A2 whole category is down	Shows the system refusing a false alarm. The SKU looks broken and is not.
3	B3 broken size curve	Stock reports look healthy and sales are falling. Impossible to see without SKU-level drill-down.
4	E1 returns spike	Gross sales fine, net sales weak. Justifies the returns table.
5	G1 unexplained	The gate opens, external evidence appears at lower confidence, and may still be ruled out. Needed or the third source is never demonstrated.

One thing to decide before writing any code
What is the baseline? The gap can be measured against the monthly target, against the same period last year, or against a trailing average. Each gives a different number and a different story. The business runs on monthly targets, so the headline gap should be against target — with the trailing average used only inside the supply checks, where the question is "how much would this have sold if it had been available".

AURELIA is a fictional company created for this case study.